



Government spying!

Governments have a very bad habit of spying. So now-a-days it is spying on our surfing data.
<http://dgit.in/WebTracked>



New bitcoin

We will soon be seeing a new Bitcoin 'Gemini' backed by a FDIC-insured bank.
<http://dgit.in/BTCGemini>

Industry Connect

the numbers but you're waiting for the data to be processed. In this regard you might have heard our CEO talk about 3D cross point technology which is revolutionary nature. Its innovation in cost per bit of a storage cell and its innovation in speed and density is great. So, we take a very systems approach to everything we do from a compute standpoint because at the end of the day the process technology is one trajectory, but you need to have a whole system balance to come along to give you the best computing experience.

Q Since you mentioned Thunderbolt, I'm curious about adoption of cutting edge technologies. When can we see mainstream adoption of things like Real Sense, WiDi, or even Thunderbolt. Is there no pull from the market, or is Intel is not pushing these technologies aggressively enough?

Anuj: No, I think it's an evolution. Like every new technology has sort of an adoption curve and the rate of adoption depends on many different factors: computing standard, how much marketing went behind it, what the ecosystem looks like. Look at Wi-Fi, circa 2003, when we packaged it as a part of the Centrino kit, it took off. Right after that you saw a high spike in notebooks. That was consequence of obviously engineering at the platform level, but more importantly I would say working within the ecosystem like hotspot vendors ensuring that for every centrino

– one way of connecting data vs another. Our approach is primarily that if you standardise, you get the scale and you can get a lot more partners to participate. But sometimes you do have people who have different views of different standards. I think as far as your Real Sense is concerned, as far as Thunderbolt is concerned, those will all have different adoption curves. It will be a function of how easy the technology is to use and how cost effective it is to use. So I can't predict exactly how fast or how slow it will go. What I can tell you is that we are working with all our partners on all those technologies and everyone is on board. Sometimes you need an operating system to come along that helps speed things up... that was Windows 10 with RealSense integrated enabling breakthrough user interface experiences such as actually unlocking your PC with your face. This is of course a step. Now it has to get more consumer awareness and utility and if that catches on it'll scale faster.

Q Regarding the whole Microsoft demo where we saw the Surface Book and other products unveiled. What kind of involvement did Intel have?

Anuj: I can tell you that Intel has collaborated very closely with Microsoft. It's been true in the past and particularly true with the launch of Windows 10 as well. So everything from speech algorithms

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there was a Wi-Fi infrastructure that we now take for granted. And the same is true for things like Thunderbolt. By virtue of it being a bleeding edge technology, it tends to be expensive when it first comes off the chute. This makes for a bit of a chicken and egg problem. You find few partners that are willing to invest in a technology that's at the bleeding edge of innovation, and over time as the waterfall happens it becomes more scalable. At the same time there are certainly competing standards

for Cortana and the DSP and processing that's required to make sure that makes a really fluid interface. Some of the data transfer and the recognition of those third party devices – those don't happen magically. Those require deep engineering collaborations from the operating system and silicon manufacturer. So yeah, whether it's the voice assistant, or it's the WiGig technology, or it's graphics optimisation, or 4K videos algorithms, there's a lot of collaborations all the way through.

Q What was the HoloLens? They say that some custom silicon HPU (Holographic Processing Unit) was used. Was that Intel or is that ARM?

Anuj: I'm not that familiar with the HoloLens specifics, we collaborated for RealSense but I don't know specifically if we did anything for HoloLens.

Q So we are witnessing tablets, convertibles and phones converging. The lines are blurring between devices and their intended purposes. We've have got PCs on a stick and NUC boxes now. As chipmakers, what is Intel's game plan in the portable space, specifically Core M and the Atom SoCs?

Anuj: So we have a roadmap for all of these. Skylake is a scalable architecture. The same microprocessor architecture can be taken all the way down to a compute stick, you can take it to a thin and light notebook, you can take it to a mobile workstation, you can take it to a gaming desktop – so that's the one architecture that spans the full gamut. There are some devices that are purpose-built and low powered. So the Atom processor line caters to those kinds of devices, typically in the, lets say 8 inches and below screens size category. Because the higher you go in screen size, the higher power it requires and you need performance as well. So think of that as we got the Atom line, we got the Core line. Now the Core M is sort of a line which is focused more on mobility, mobile computing, so you think of the thinnest and lightest notebooks where mobility is an everyday concern. But if you want performance and you don't want that hourglass to happen when you're video editing, then you can have a Core i-series.

Think of it as one being more mobile focused, one more performance focused, and within those you would have good, better and best segments. So it's Core M3, Core M5, Core M7 and Core i3, i5 and i7. So that segmentation has worked really well for us. Even in the end user's mind also, they first have to make a simple decision like do you want an exclusively performance oriented device that pushes the envelope, or do you want a device that's a lot more mobile. Once you make those decisions you go with a good, better and the best. **Q**